CYGNUS-A2 APPLIED RESEARCH

THAL3S16 Meta-Model Financial Intelligence

Institutional Research Report — Version 9

Cygnus-A2 Applied Research — THAL3S16 Program
Version 9 (V9) — research freeze 2026-11-01
Narrative source: `docs/THAL3S16-RESEARCH-MANUSCRIPT-V9.md` (SprintVis S04)
Charts: research chart set v1 (SprintVis S05), embedded vector via svglib
Toolchain: reportlab 5.1.0 + svglib 2.3.2 (hash-locked), base-14 fonts

CONFIDENTIALITY NOTICE — Private research artifact. Prepared by Cygnus-A2 for internal use and authorized reviewers only; not for public distribution. No confidential founder policy text and no licensed ATTOM data is reproduced or embedded anywhere in this document; only derived, committed, audited records appear.

RESEARCH STATUS NOTICE — This is a research report, not a product claim. No statistical significance, model-versus-model superiority, autonomous-lending or funding-capability claim is made or supported: quantities that are not measured are labeled NOT_MEASURED / UNKNOWN / UNDETERMINED and are never rendered as zero. Every decision record in the underlying research carries approval_authorized=false, issuance_authorized=false, funding_authorized=false and human_review_required=true.

DATA LABEL — All client entities and property records in this report are fictional: SYNTHETIC_AUTHORED_ILLUSTRATIVE_ONLY_NOT_REAL_CLIENT_DATA. Nothing in this document is credit advice, an approval, an issuance or a funding commitment.

About This Document

Document status: Research manuscript (V9), edited from the committed V8 research report content. Narrative source for the V9 PDF. Every measured figure in this manuscript is read back programmatically from the committed artifacts named in the provenance and digest registers (Appendices C–D) and is anchored by the committed suite `demo/tests/test_sprintvis_s04_manuscript.py`. Nothing is hand-transcribed; anything not computed from a committed record stays NOT_MEASURED, UNKNOWN or UNDETERMINED — never zero, never invented.

Scope and honesty pins (carried from the V8 report, unchanged). All client entities in this research are fictional and labeled SYNTHETIC_AUTHORED_ILLUSTRATIVE_ONLY_NOT_REAL_CLIENT_DATA. No output is credit approval, issuance or funding: every decision record carries approval_authorized=false, issuance_authorized=false, funding_authorized=false and human_review_required=true. Draft policy thresholds are labeled DRAFT_NOT_PRODUCTION_APPROVED / PROPOSED_UNCALIBRATED / PENDING_POLICY_SOURCE and are never presented as production-approved. No significance claim, no model-versus-model quality ranking, no autonomous-lending capability and no liquidity commitment is claimed anywhere in this manuscript: the live evidence base does not support any of them, and the sections below say exactly what would be required before such statements could ever be made.

CONFIDENTIALITY NOTICE — Private research artifact. Prepared by Cygnus-A2 for internal use and authorized reviewers only; not for public distribution. No confidential founder policy text and no licensed ATTOM data is reproduced or embedded anywhere in this document; only derived, committed, audited records appear.

RESEARCH STATUS NOTICE — This is a research report, not a product claim. No statistical significance, model-versus-model superiority, autonomous-lending or funding-capability claim is made or supported: quantities that are not measured are labeled NOT_MEASURED / UNKNOWN / UNDETERMINED and are never rendered as zero. Every decision record in the underlying research carries approval_authorized=false, issuance_authorized=false, funding_authorized=false and human_review_required=true.

DATA LABEL — All client entities and property records in this report are fictional: SYNTHETIC_AUTHORED_ILLUSTRATIVE_ONLY_NOT_REAL_CLIENT_DATA. Nothing in this document is credit advice, an approval, an issuance or a funding commitment.

Contents

About This Document . . . 2

Style and Terminology Key (applies to this entire manuscript) . . . 5

1. Executive Summary . . . 6

2. Research Motivation and the Cygnus-A2 Value Proposition . . . 7

2.1 The problem . . . 7

2.2 The approach . . . 7

2.3 Why this matters commercially . . . 8

3. System and Methodology . . . 8

3.1 The decision engine (deterministic core) . . . 8

3.2 Credit-policy coverage audit . . . 9

3.3 Benchmark populations (authored, frozen, no model calls) . . . 10

3.4 The 30-client demonstration portfolio . . . 11

3.5 The recovered property-intelligence frontend . . . 12

3.6 The meta-model layer . . . 12

4. Evidence and Results . . . 13

4.1 Deterministic reference assessments (corpus honesty) . . . 13

4.2 30-client integrated outcomes (recomputed) . . . 13

4.3 Live inference record (the honest denominator) . . . 16

4.4 Financial grounding and accuracy observations . . . 18

4.5 Latency and token efficiency (committed receipts only) . . . 19

4.6 Cost position and envelopes (integer micro-USD; ESTIMATED) . . . 20

4.7 Safety and failure attribution . . . 22

5. Statistical Adequacy of the Evidence Base . . . 23

5.1 What the 384 cases are — and are not . . . 24

5.2 Effective independent sample size (computed bounds) . . . 25

5.3 Required independent case counts (Wilson 95%) . . . 25

5.4 Paired-comparison power (matched cases) . . . 25
5.5 Per-regime adequacy . . . 26
5.6 Rare safety-failure detection limits . . . 26
5.7 Live evidence position and honest intervals . . . 27

6. Model Comparison Readiness . . . 27

7. Limitations and Honest-Statement Registry . . . 28

8. Proposed Next Steps (all labeled Proposed: — none is an executed result) . . . 29

Appendix A — Machine-Readable Identifier Register . . . 30

Appendix B — Provenance Register (source commits) . . . 32

Appendix C — Digest Register (sha256, recomputed at authoring) . . . 33

Appendix D — Procedural Commentary (sprint mechanics; kept out of the narrative) . . . 33

Appendix E — Glossary for Non-Engineering Readers . . . 34

Style and Terminology Key (applies to this entire manuscript)

- **Headings:** numbered sections, Title Case; table headers sentence case.
- **Evidence buckets (verbatim, never blended):** LIVE_RECEIPT (committed

sanitized receipts + used-markers, metadata only), OFFLINE_MOCK (scripted mock runs and authored harness constants; never model output), RECORDED (committed replays/walkthroughs; observations), SYNTHETIC (authored fixtures and corpora; never real client data), INCOMPLETE (never-validated attempts and approvals without receipt; counted, never interpreted as quality). The legacy label REAL_INFERENCE denotes the same bucket as LIVE_RECEIPT.

- **Epistemic markers:** statements are labeled inline with **Evidence:**

(read back from a committed record), **Interpretation:** (editorial reading of evidence, not itself evidence), **Hypothesis:** (untested proposal about mechanism) and **Proposed:** (a proposed improvement, not implemented results). Interpretations and proposals are never presented as findings.

- **Money:** canonical values are integer USD cents or integer micro-USD

(1 USD = 1,000,000 micro-USD); USD display strings appear only at this presentation boundary and are computed from the integers. Missing money is null + an explicit status, never zero. Costs are ESTIMATED-labeled; actual billed amounts are UNKNOWN.

- **Units:** basis points are "bps" (1 bps = 0.01 percentage point);

percentage-point differences are "pp"; latencies are integer nanoseconds (ns) canonically, with seconds shown for readability.

- **Decision statuses:** the narrative uses readable labels — "indicative

capacity" [ILLUSTRATIVE_CAPACITY], "review required" [REVIEW_REQUIRED], "blocked" [BLOCKED]. The machine-readable tokens (uppercase, exact) are retained in the technical appendices and in every table, because they are the committed contract vocabulary. BLOCKED implies capacity 0.

- **Missing observations:** NOT_MEASURED / UNKNOWN / UNDETERMINED stay that

way; they are never zero-filled and never converted into claims.

- **Planning values:** assumed planning rates are labeled

PLANNING_ASSUMPTION — they are inputs to sample-size arithmetic, never measurements.

- **Confidentiality:** the founder's confidential credit-policy documents are

never reproduced; only the derived, committed, audited coverage matrix and its public descriptions appear here. Licensed ATTOM values are not embedded anywhere. The original demo property is referenced by role, never by its address literal.

1. Executive Summary

THAL3S16 is a private-research credit-intelligence platform for ultra-high-net-worth (UHNW) property-backed lending, built by Cygnus-A2 around one governing idea: **a lending decision aid must be auditable end to end, fail closed, and honest about what it has and has not measured.** This manuscript is the integrated research record of the platform as it stands today — what was built, what the committed evidence shows, what the evidence does not yet show, and what would be required to close the gap.

What the platform is. A deterministic decision engine (credit-engine-v2) reads a borrower's evidence — property valuations, secured debt, liquid assets, cash flow, risk dimensions — under a frozen, versioned contract and produces one of three outputs: an indicative capacity [ILLUSTRATIVE_CAPACITY], a mandatory human review [REVIEW_REQUIRED], or a block [BLOCKED]. The engine uses exact integer arithmetic (cents and basis points; no floats in the money chain), a founder-approved conservative ceiling rounding convention (ltv-rounding-contract-v2), and authority pins on every record: nothing is ever approved, issued or funded by the system, and human review is always required. A meta-model layer (external language models behind a frozen validator) is being evaluated for its ability to read and ground the underlying evidence; a recovered public web frontend demonstrates the engine's outputs for thirty fictional clients on an interactive California map.

What the committed evidence shows.

- **Evidence:** The founder's confidential credit-policy documents were

audited offline; the committed derived matrix records **57 rules** (32 from Document A, 25 from Document B) with all 57 page anchors verified, and coverage of 15 IMPLEMENTED_GROUNDED, 14 PARTIALLY_IMPLEMENTED, 5 PROPOSED_UNCALIBRATED and 23 NO_MAPPING. The founder's stated "57 audited rules" figure was verified against the committed record and matches (`rules_total: 57` in `policy_coverage_matrix_v1.json`).

- **Evidence:** Across the 30-client fictional demonstration portfolio, the

engine's recomputed decisions (mismatches vs the frozen reference assessments: 0 of 30) yield 9 indicative-capacity outcomes, 8 review-required outcomes and 13 blocked outcomes, with a total indicative capacity of 10,221,500,000 cents (USD 102,215,000.00) — every figure recomputed in-process, never hand-copied.

- **Evidence:** Two deterministic benchmark corpora — a 96-case packet and a

384-case balanced factorial census (seed 20261031) — re-derive with zero invariant failures: 0 of 96 and 0 of 384 checks mismatch. These corpora are authored synthetic populations, not live results: completed live assessments of the 384-case benchmark number **zero**.

- **Evidence:** The honest live evidence position is small and is reported

without inflation: **3 completed live model assessments** (all single, non-benchmark cases on the one execution-path-tested route), 15 committed receipts against 17 approvals, and a cumulative estimated cost position of **128,950 micro-USD (USD 0.128950)** over committed records — ESTIMATED, never billed. Live attempt registers record 10 attempted meta-assessment cases with 2 COMPLETED on the fixture's own counting convention, and the private ledger census records 9 attempt-classified receipts with 3 COMPLETED on its convention; both registers are reported exactly as each record states them and are never blended.

- **Evidence:** The recovered frontend is byte-identical to the founder's

live public deployment (zero drift across all four public files), carries 35 map markers (30 fictional clients + the 5 original demo properties), and matched the engine's records to the cent for all 30 clients in a recorded browser walkthrough.

What the evidence does not show — stated plainly. No model-quality rate, interval or ranking exists for any benchmark population (NOT_MEASURED). Live safety behavior is essentially unmeasured (zero failures on n=3 completed calls; the computed 95% upper bound on the failure rate is 56.2%). Actual billed costs are UNKNOWN. The defensible effective independent sample size of the 384-case census is between 16 and 96 units — not 384 — and every pre-registered minimum for a statistically adequate live evaluation (60 independent completed cases for a ±10 pp interval at the planning rate; 118 completed paired cases for a 10 pp paired minimum detectable difference; 73 completed calls to bound the safety-failure rate at 5%) is unmet by one to two orders of magnitude today.

Interpretation: The platform's contribution at this stage is not a measured model capability; it is the *auditable infrastructure* — frozen contracts, deterministic recomputation, evidence buckets, pre-registered statistics and refusal semantics (the comparison module refuses to declare a winner by design) — that makes any future live claim checkable. That is the Cygnus-A2 value proposition: a credit-research platform in which every number a judge or investor sees can be traced to a committed record, and in which the honest boundaries of the evidence are part of the product.

Proposed: The next milestone (not yet executed; requires founder authorization and case-generation beyond the current fixture) is a live paired evaluation of at least 118 completed matched cases against the pre-registered criteria in Section 6 — the smallest pre-registered design that could support any comparative statement.

2. Research Motivation and the Cygnus-A2 Value Proposition

2.1 The problem

UHNW property-backed lending decisions are evidence-heavy and evidence-fragile. A single file can contain several valuations of the same property that disagree; valuations age at different rates; ownership and lien records can conflict; borrower liquidity is spread across assets whose realizability differs; and risk concentrates across interacting dimensions — concentration that a single-number score smooths over. Human credit committees read this evidence well but slowly, inconsistently, and without a shared audit trail.

The industry's two default answers both fail this setting. Opaque automated underwriting is unacceptable for bespoke credit at this ticket size and is increasingly untenable regulator- and reputation-wise. Unstructured analyst memo drafting leaves no reproducible decision path and no way to re-derive yesterday's answer. The missing artifact is a *disciplined evidence-reading layer*: deterministic, versioned, and honest about uncertainty.

2.2 The approach

Cygnus-A2's THAL3S16 platform is that layer, built under four non-negotiable disciplines:

1. **Deterministic core, exact arithmetic.** The decision engine is a frozen,

versioned contract (credit-engine-v2; request schema credit-request-v1; rounding contract ltv-rounding-contract-v2). Money is integer cents; ratios are integer basis points computed under a founder-approved conservative ceiling convention; no float enters any money chain. Regeneration is

byte-reproducible, so any figure can be recomputed and any discrepancy is a defect, not a rounding story.

2. **Fail-closed semantics.** Missing evidence blocks or forces review; it

never silently zeroes. Authority pins on every record read approval_authorized=false, issuance_authorized=false, funding_authorized=false, human_review_required=true. The engine can render an indicative capacity; it can never authorize anything.

3. **Evidence buckets, never blended.** Every figure in the research corpus

carries one of five evidence labels — LIVE_RECEIPT, OFFLINE_MOCK, RECORDED, SYNTHETIC, INCOMPLETE — and a figure may never cross buckets. Scripted harness runs validate machinery; authored fixtures define expectations; only committed live receipts count as live evidence. A rejected attempt is counted, never graded.

4. **Pre-registered statistics.** Sample-size requirements, minimum

detectable differences, multiplicity controls and the criteria under which any model-versus-model statement could ever be made were computed and committed *before* any live batch, and the comparison module refuses to publish when its evidence bar is unmet — which it currently is.

2.3 Why this matters commercially

Interpretation: A lending platform whose every displayed figure recomputes from committed records, whose draft thresholds are labeled as drafts, and whose limitations section is written by the statistics module rather than the marketing function is structurally cheaper to diligence — by a founder, an investor, a judge or a regulator — than any alternative that asks the reader to trust prose. The value of THAL3S16 at this stage is precisely that it does not oversell: what remains unmeasured is on the page. The follow-on value is speed with auditability — when live evaluation is authorized, the same frozen contracts and pre-registered criteria convert raw model calls into decision-grade evidence without a new methodology sprint.

3. System and Methodology

3.1 The decision engine (deterministic core)

The canonical engine (`demo/credit_engine.py`, credit-engine-v2) is a pure, total function over a versioned request schema (credit-request-v1). Its decision chain, in order: collateral eligibility -> property-evidence usability (fresh/consistent evidence required) -> financial evidence presence -> LTV chain (valuation spread and staleness haircuts, capped at 1500 bps, applied to the 4000 bps illustrative LTV anchor) -> per-property capacity contributions (value × effective LTV - property debt - costs, floored at zero) -> client-level aggregation (other liabilities subtracted once) -> risk dimension triggers -> concentration review. The output vocabulary is exactly three statuses — ILLUSTRATIVE_CAPACITY, REVIEW_REQUIRED, BLOCKED — and BLOCKED implies capacity 0. Every numeric threshold inside the chain remains PROPOSED_UNCALIBRATED / PENDING_POLICY_SOURCE except the founder-approved v2 calibration described in Section 3.2.

For readers without a software background: the engine is a fixed mathematical pipeline, like a building code — same inputs, same outputs, every time, forever — and this manuscript's figures were produced by re-running that pipeline, not by reading old reports.

Worked example (SYNTHETIC_DETERMINISTIC, computed in-process from the DEMO_ block, never transcribed). For the illustrative demo parameters — value 500,000,000 cents (USD 5,000,000.00), requested 150,000,000 cents (USD 1,500,000.00), existing secured debt 50,000,000 cents, reserved costs 10,000,000 cents: the requested LTV is the exact ceiling of 150,000,000 × 10,000 / 500,000,000 = **3000 bps** (ltv-rounding-contract-v2, ceiling); the effective LTV is **4000 bps** (haircut 0); the borrowing-limit ceiling is 500,000,000 × 4000 / 10,000 = **200,000,000 cents** (USD 2,000,000.00); and the indicative capacity after debt and costs is **140,000,000 cents (USD 1,400,000.00)**. All integer arithmetic.

The DEMO_ parameters themselves (illustrative LTV anchor 4000 bps; worked value 500,000,000 cents; worked request 150,000,000 cents; staleness haircut 2500 bps; per-regime stress haircuts) are illustrative demo constants labeled PROPOSED_UNCALIBRATED / PENDING_POLICY_SOURCE — they are NOT credit policy.

3.2 Credit-policy coverage audit

The founder supplied two confidential credit-policy documents (Document A, the external policy; Document B, the framework draft) via a private route. They were audited offline; the documents themselves are never committed or reproduced. The derived, committed coverage matrix (`policy_coverage_matrix_v1.json`, frozen audit-time record) records:

Audit fact (read from the committed matrix)	Value
Rules extracted (total)	57
— from Document A	32
— from Document B	25
Page anchors verified	57 of 57 (anchor failures: 0)
IMPLEMENTED_GROUNDED	15
PARTIALLY_IMPLEMENTED	14
PROPOSED_UNCALIBRATED	5
NO_MAPPING	23

By pillar: Multi-Dimensional Risk Intelligence 37, Borrower Financial Capacity 6, Collateral Assessment 7, Cross-Pillar Governance 7. All 25 Document B rules carry document maturity DRAFT_INTERNAL_NOT_PRODUCTION_APPROVED: **no draft item is presented as production-approved anywhere in this research.**

The single applied calibration (credit-engine-v2, founder-approved bounded GO) is the concentration-response review branch: when at least DEMO_RISK_CONCENTRATION_HIGH_DIMENSIONS = 3 applicable HIGH risk dimensions are present, the outcome changes from blocked to review-required (decline-or-review), the capacity is still rendered, and mandatory human review is flagged — never auto-approved. Its policy rule references are POL-A-027, POL-B-019 and POL-B-022; its maturity is GROUNDED_SIGNAL_PROPOSED_RESPONSE (a grounded signal with a proposed, uncalibrated response). The pre-calibration semantics (early block, capacity zero, reason BLOCKED_RISK_CONCENTRATION) are preserved as the recorded historical behavior; no historical output was rewritten.

Evidence: the matrix is an audit-time snapshot (engine_version credit-engine-v1 at audit commit) sitting beside a live credit-engine-v2; the audit-time citations still resolve on the live engine, and the two calibrated clients differ only in their review reason code — a recorded, test-pinned difference, not a discrepancy.



Figure 1. Credit policy implementation coverage — audited rule count — n = 57 audited policy rules. Units: rule count (categorical audit status; no score blending). Sources: demo/fixtures/policy_coverage_matrix_v1.json (label: DERIVED_FROM_FOUNDER_CONFIDENTIAL_POLICY_PRIVATE_REPO_ONLY); anchors verified: 57/57 (extraction_integrity). Data label: DERIVED_FROM_FOUNDER_CONFIDENTIAL_POLICY_PRIVATE_REPO_ONLY.

3.3 Benchmark populations (authored, frozen, no model calls)

Two deterministic corpora define expected engine behavior:

- **96-case packet** (meta_benchmark_96_v1, seed 20261030): 96 cases — 54

blocked, 42 illustrative-only; blocking reasons REQUEST_EXCEEDS_CAPACITY 30, STALE_EVIDENCE 24, VALUE_CONFLICT 24.

- **384-case census** (meta_benchmark_384_v1, seed 20261031, status

FROZEN_PREPARATION_NO_MODEL_CALLS): a fully balanced factorial design — 4 UHNW client profiles × 4 portfolio configurations × 6 economic regimes × 4 authored variants = **384** case records; 96 base scenarios × 4 cases each; 16 financial situations × 24 cases each; expected outcomes 114 ILLUSTRATIVE_ONLY / 270 BLOCKED, identical (19/45) in every regime; 96 variant-0 cases are byte-twins of the 96-case packet; 0 exact duplicate fact projections (384 unique digests); stage subsets of 48 and 96 cases; authorization flags true: 0; human_review_required true: 384.

Economic regimes (six, from the committed macro fixtures): normal, high_interest_rates, property_downturn, liquidity_crisis, recession, recovery — four backed by frozen official macro snapshots, two authored synthetic and labeled AUTHORED_SYNTHETIC_NOT_OBSERVED. A separate 13-regime scenario library (from the

frozen `demo/economic_scenario_results.json`) exists for harness stress notation; regime identifiers are verbatim and never renamed.

Both corpora are SYNTHETIC: authored balanced factorial censuses, not live model assessments and not a representative sample of real UHNW clients. Every case is ILLUSTRATIVE_ONLY with authorization flags false and human review required — the corpora measure evidence reading, never lending.

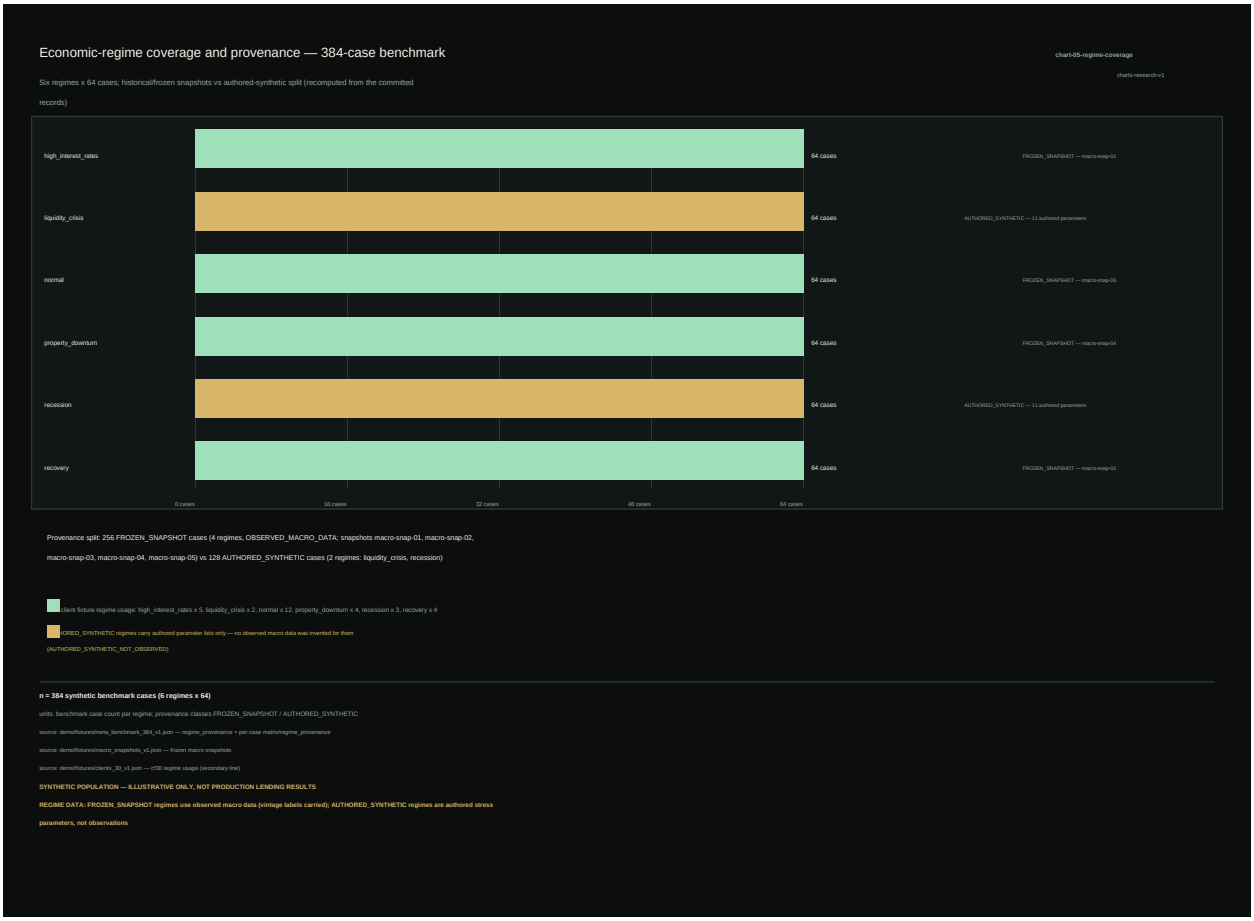


Figure 2. Economic-regime coverage and provenance — 384-case benchmark — n = 384 synthetic benchmark cases (6 regimes x 64). Units: benchmark case count per regime; provenance classes FROZEN_SNAPSHOT / AUTHORED_SYNTHETIC. Sources: demo/fixtures/meta_benchmark_384_v1.json — regime_provenance + per-case matrix/regime_provenance; demo/fixtures/macro_snapshots_v1.json — frozen macro snapshots; demo/fixtures/clients_30_v1.json — cf30 regime usage (secondary line). Data label: SYNTHETIC_AUTHORED_ILLUSTRATIVE_ONLY_NOT_REAL_CLIENT_DATA.

3.4 The 30-client demonstration portfolio

Thirty fictional clients (cf30-001 … cf30-030, schema clients30-v1, data label SYNTHETIC_AUTHORED_ILLUSTRATIVE_ONLY_NOT_REAL_CLIENT_DATA) span client categories A–E with per-client properties, evidence states, debt, liquid assets, cash flow and allocation profiles. The fixture embeds a deterministic reference assessment per client (label SYNTHETIC_DETERMINISTIC), and every figure reported in Section 4.2 is recomputed through the canonical engine and compared field-by-field against those references before rendering.

3.5 The recovered property-intelligence frontend

The public demonstration frontend was recovered via read-only HTTPS from the founder's own deployment (thal3s16-property-preview.pages.dev, deployment id aec7a228-b215-4a68-8f87-8b3aa2718925, recovered 2026-11-01): four served files, reconstructed from publicly served bytes only. A committed pinned-diff contract enforces no-redesign: the integrated JavaScript differs from the pristine recovery by exactly three removed original lines plus one clearly marked additive integration block, and the page adds exactly one script tag. The map carries 35 markers: the 30 fictional clients plus the 5 ORIGINAL demo properties (Beverly Hills, Malibu, Montecito, Palo Alto, La Jolla), which remain unchanged. The original demo property is referenced by role only; its address literal is confined to the three recovered files (anti-spread pin) and appears nowhere in this manuscript.

Evidence: a recorded browser QA walkthrough (Sprint-Orig S07, committed report) selected all 30 clients, ran each seven-stage assessment, and matched the DOM to the records **to the cent for all 30 clients**; the committed recovery is byte-identical to the live deployment (zero drift across 4 of 4 public files); secret-shape and policy n-gram scans over the served files recorded zero hits.

3.6 The meta-model layer

The meta-model layer is the part of the platform that evaluates whether external language models can read credit evidence accurately under a frozen contract. Its components:

- **Access roster** (five models, frozen configuration, no model calls):

one execution-path-tested route (muse_spark_open via Fireworks hosting); the other four models are not available on a viable route at audit time (two blocked on absent credentials, one publicly listed but route-blocked, one not served on a viable route). All live calls in the record are on the tested route.

- **Frozen validator:** 13 fail-closed rejection classes (authority

override, missing/extra fields, wrong schema, wrong model, stripped provenance, invented latency, flipped safeguards, unsupported numeric claims, ungrounded interpretation, unknown fact references, false abstention, forged grounding, invalid config) — a response that fails any class is rejected and never graded.

- **Compact reasoning contract:** a token-budget layer over the meta

assessment prompt with deterministic trimming; all its comparison results are simulated OFFLINE_MOCK (zero genuine receipts exist for it; live capability UNKNOWN until separately founder-authorized).

- **Comparison-preparation module** (`demo/meta_comparison_prep_v1.py`):

a matched-case planning framework with six separated dimensions (financial grounding, safety, latency, cost, abstention, completed throughput), publish gates that refuse when a dimension is unmeasured, and a pre-registered winner evidence bar (Section 6).

For non-engineers: the "meta layer" is a testing laboratory for language models. The models never touch money or decisions; they are graded on how faithfully they read evidence, under rules strict enough that a model cannot pass by sounding confident.

4. Evidence and Results

4.1 Deterministic reference assessments (corpus honesty)

Every case of both corpora re-derives from its own fact projection with zero invariant failures: **96 of 96 checked, 0 failures; 384 of 384 checked, 0 failures** (label SYNTHETIC_DETERMINISTIC, expectations label DETERMINISTIC_EXPECTATIONS). The 384-case blocked reasons: REQUEST_EXCEEDS_CAPACITY 186, STALE_EVIDENCE 120, VALUE_CONFLICT 120.

Interpretation: zero invariant failures means *corpus honesty* — the authored expectations match what the frozen engine actually derives. It is a statement about the test fixture and the engine agreeing with each other; it is NOT a statement about model quality, which is NOT_MEASURED on these populations.

4.2 30-client integrated outcomes (recomputed)

All 30 clients were re-assessed through the canonical engine and compared field-by-field (status, capacity, reason codes, requested LTV) against the fixture's embedded references: **mismatches 0 of 30.**

Outcome (readable label)	Machine status	Clients	Indicative capacity (cents, sum)
Indicative capacity	ILLUSTRATIVE_CAPACITY	9	7,229,500,000 (USD 72,295,000.00)
Review required	REVIEW_REQUIRED	8	2,992,000,000 (USD 29,920,000.00)
Blocked	BLOCKED	13	0

Total indicative capacity over the 30 clients: **10,221,500,000 cents (USD 102,215,000.00)**, accumulated by explicit integer summation. The reason-code census (recomputed): BLOCKED_REQUEST_EXCEEDS_CAPACITY 6, BLOCKED_OWNERSHIP_OR_LIEN_EVIDENCE 3, BLOCKED_INSUFFICIENT_PROPERTY_EVIDENCE 2, BLOCKED_INELIGIBLE_COLLATERAL 1, BLOCKED_MISSING_FINANCIAL_EVIDENCE 1, REVIEW_INSUFFICIENT_CASHFLOW 5, REVIEW_RISK_CONCENTRATION 2, REVIEW_SEVERE_SINGLE_DIMENSION 1.

Authority pins recounted on every recomputed decision: approval_authorized=false 30/30, issuance_authorized=false 30/30, funding_authorized=false 30/30, human_review_required=true 30/30; threshold_calibration PENDING_POLICY_SOURCE on 30/30 records.

Frontend parity (verified in-process): the generated judge-flow payload (r12-judge-flow-v1, byte-reproducible) carries capacity-state counts INDICATIVE 9 / REVIEW_REQUIRED 8 / BLOCKED 13 under the documented mapping ILLUSTRATIVE_CAPACITY -> INDICATIVE, and every per-client indicative capacity equals the recomputed engine capacity (30/30); a mismatch aborts rendering. The recorded browser QA distribution (INDICATIVE 9 / REVIEW_REQUIRED 8 / BLOCKED 13) agrees three ways.

Evidence label: SYNTHETIC_DETERMINISTIC — deterministic reference assessments over fictional clients; corpus honesty, never a lending decision. Live model behavior on any client profile is NOT_MEASURED.

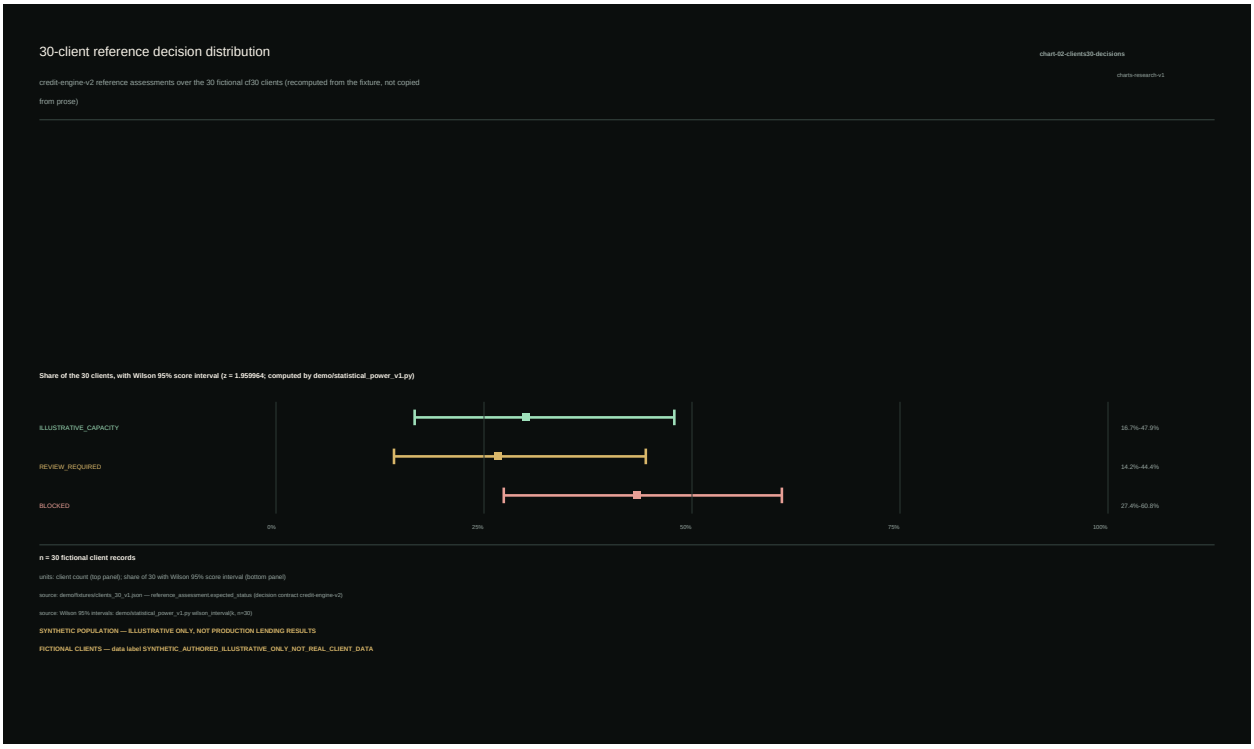


Figure 3. 30-client reference decision distribution — n = 30 fictional client records. Units: client count (top panel); share of 30 with Wilson 95% score interval (bottom panel). Sources: demo/fixtures/clients_30_v1.json — reference_assessment.expected_status (decision contract credit-engine-v2); Wilson 95% intervals: demo/statistical_power_v1.py wilson_interval(k, n=30). Data label: SYNTHETIC_AUTHORED_ILLUSTRATIVE_ONLY_NOT_REAL_CLIENT_DATA.

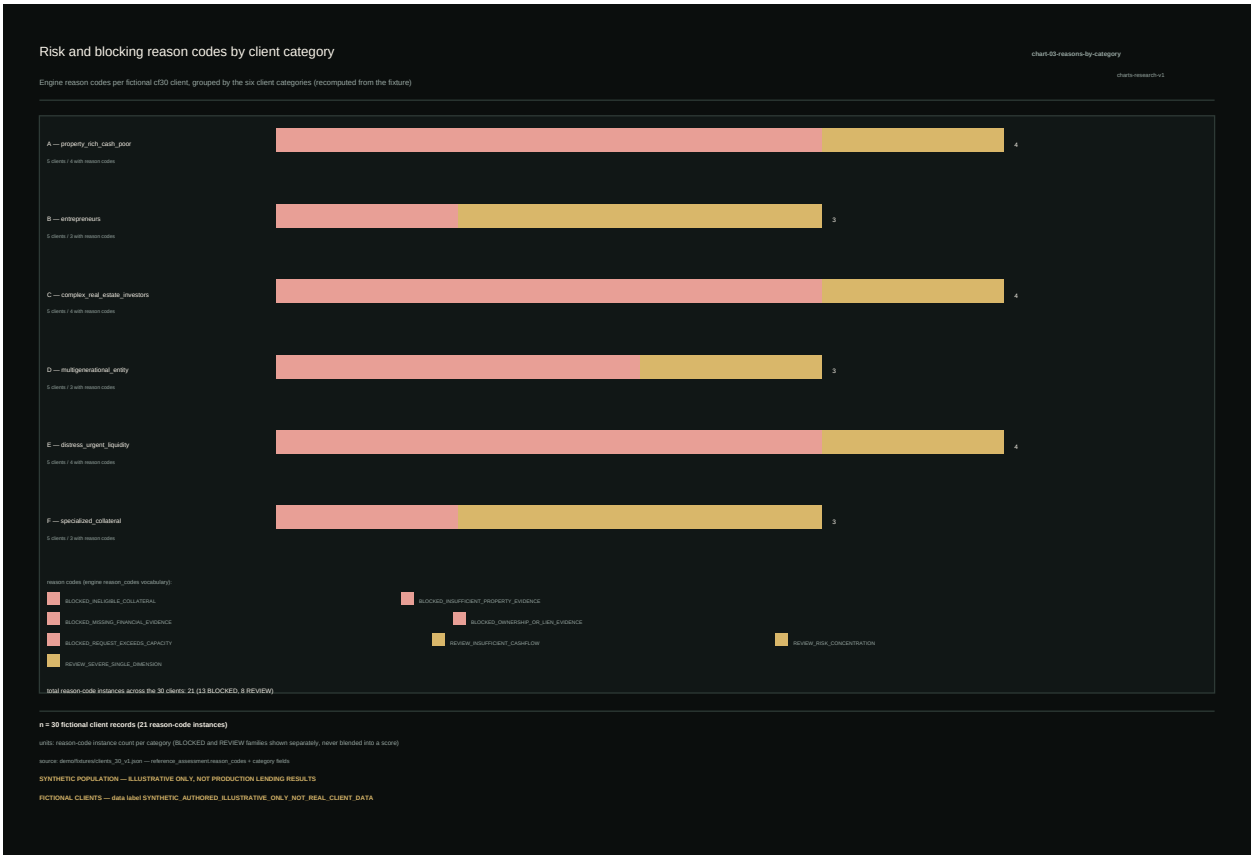


Figure 4. Risk and blocking reason codes by client category — n = 30 fictional client records (21 reason-code instances). Units: reason-code instance count per category (BLOCKED and REVIEW families shown separately, never blended into a score). Sources: demo/fixtures/clients_30_v1.json — reference_assessment.reason_codes + category fields. Data label: SYNTHETIC_AUTHORED_ILLUSTRATIVE_ONLY_NOT_REAL_CLIENT_DATA.

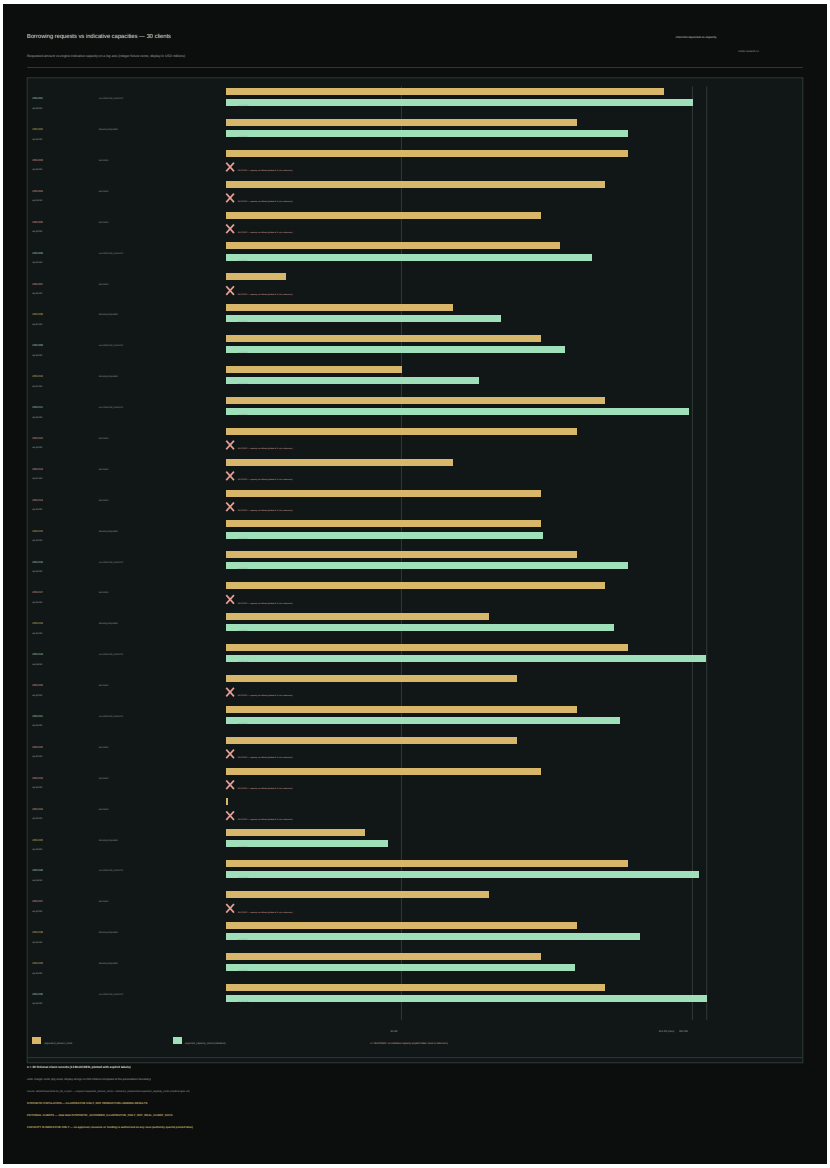


Figure 5. Borrowing requests vs indicative capacities — 30 clients — n = 30 fictional client records (13 BLOCKED, plotted with explicit labels). Units: integer cents (log scale; display strings in USD millions computed at the presentation boundary). Sources: demo/fixtures/clients_30_v1.json — request.requested_amount_cents + reference_assessment.expected_capacity_cents (credit-engine-v2). Data label: SYNTHETIC_AUTHORED_ILLUSTRATIVE_ONLY_NOT_REAL_CLIENT_DATA.

4.3 Live inference record (the honest denominator)

Live evidence is counted only from committed receipts and used-markers (metadata only). Two committed registers state it, each on its own counting convention — reported separately, never blended:

Register A — meta-assessment attempt register (committed `sprint05_statistics_v3.json`): 10 attempted meta-assessment cases; 2 COMPLETED (r11-single-call:valid; meta_v2_canary_low_reasoning); 8 REJECTED_VALIDATION. Fact accuracy was SCORED for 1 case and NOT_SCORED for 9 cases rejected before grading (a rejection is never reported as an observed 0%).

Register B — private ledger census (read-only enumeration): 15 receipts / 15 used-markers / 17 approvals (2 approvals without receipt — INCOMPLETE bucket, existence counted only); 9 attempt-classified meta

single-call receipts (3 COMPLETED, 6 REJECTED_VALIDATION) plus 6 verification calls on a separate schema; estimated cost total 101,921 micro-USD (ESTIMATED); the committed stage-1 record separately carries MEASURED spend of 54,993 micro-USD with 0 of 21 packet rows validated.

The honest completed-live count is 3 model assessments — the single property-evidence case, one same-case repeat call, and one low-reasoning canary — all on the execution-path-tested route, all single non-benchmark cases. **Completed live assessments of the 384-case benchmark: 0** (the only attempts, mb96-000 and mb384-005, were REJECTED_VALIDATION and are NOT_SCORED). Every other roster model has zero live calls.

The one fully SCORED live meta case (n=1, frozen grader): factual grounding 5 of 5, explanation completeness 5 of 5, review-priority recall 2 of 2, evidence-gap recall 2 of 2, unsupported numeric claims 0, safety preserved true; semantic grounding SEMANTIC_GROUNDING_NOT_ESTABLISHED. A single observation with no interval — reported as exactly that.

Evidence labels: LIVE_RECEIPT for the receipts and used-markers; INCOMPLETE for never-validated attempts and approvals without receipts. **Interpretation:** the live record to date is a proof of *route and machinery* — transport, validation and grading all work end to end — and explicitly not a measurement of model capability.

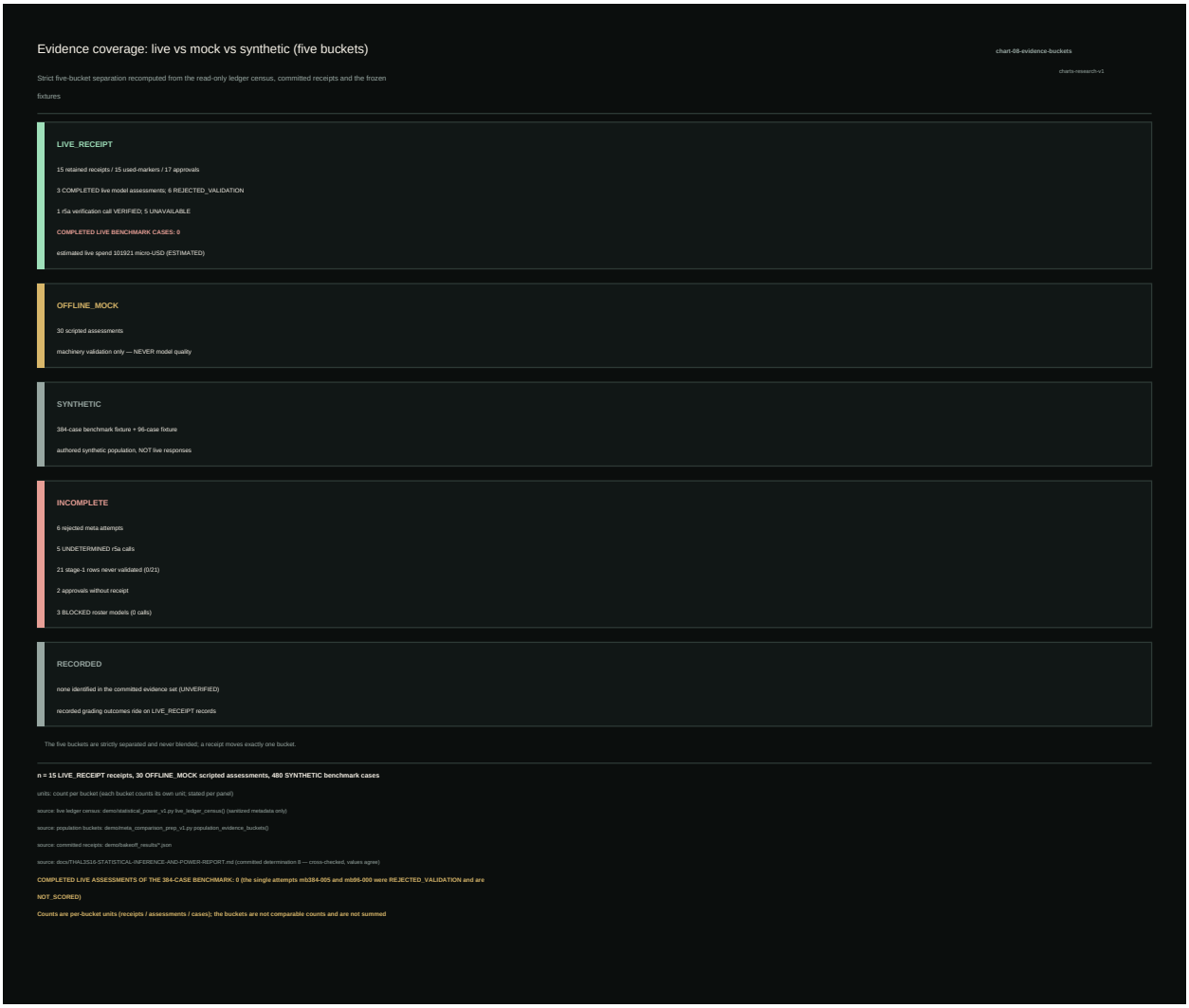


Figure 6. Evidence coverage: live vs mock vs synthetic (five buckets) — n = 15 LIVE_RECEIPT receipts, 30 OFFLINE_MOCK scripted assessments, 480 SYNTHETIC benchmark cases. Units: count per bucket (each bucket counts its own unit; stated per panel). Sources: live ledger census: demo/statistical_power_v1.py live_ledger_census() (sanitized metadata only); population buckets: demo/meta_comparison_prep_v1.py population_evidence_buckets(); committed receipts: demo/bakeoff_results/*.json; docs/THAL3S16-STATISTICAL-INFERENCE-AND-POWER-REPORT.md (committed determination 8 — cross-checked, values agree).

4.4 Financial grounding and accuracy observations

- **Property meta experiment** (LIVE_RECEIPT, n=1): expected-fact coverage

8 of 12; numeric claims 4, all 4 grounded; frozen-grader factual grounding 5 of 5 and explanation completeness 5 of 5; semantic grounding SEMANTIC_GROUNDING_NOT_ESTABLISHED. Miss attribution (verbatim statuses): the 2 uncited expected facts are DETERMINABLE_FROM_RETAINED_SCORING_METADATA; the 2 cited-but-value-missed facts are DETERMINABLE_BY_ELIMINATION_FROM_RETAINED_SCORING_METADATA; the model's wording for each miss is UNDETERMINED_FROM_RETAINED_EVIDENCE (the answer text was never retained).

- **Numeric grounding evaluation** (authored synthetic, OFFLINE_MOCK): 11

challenge examples, 46 number occurrences, 38 correct renderings; unit errors detected 2 / missed 0; attribution-error occurrences 2; unsupported-claim occurrences 1; human-review routing 2 of 46. Status: DISABLED EVALUATION PROTOTYPE — never imported by any execution, gate or runner path; no activation flag exists; no auto-acceptance anywhere.

- **Evaluator research corpus** (OFFLINE_MOCK, research path only,

authorization impact NONE): 10 authored examples; attribution classes — model failure 5, validator failure 1, schema failure 1, evidence failure 0, undetermined 2, no failure 1. Separate research analysis path; no outcome can authorize funding, approve an assessment, override a safety rejection or change any stored result.

- **Harness validation** (OFFLINE_MOCK, scripted provider): 30 scripted

evaluations, all 30 validated, across 5 asset types and 13 economic scenarios — machinery validation, not model quality; the harness latency fields (0.0) are placeholders, NOT measurements.

Honest summary: the only real grounding receipts are single observations (n=1); no cross-property, repeatability or cohort-level accuracy claim is made. Repeatability: 2 of 3 planned same-case calls are recorded (1 completed, 1 rejected); pairwise agreement NOT_MEASURED.

4.5 Latency and token efficiency (committed receipts only)

Latency appears in committed records only as integer nanoseconds from receipt fields (LIVE_RECEIPT provenance). Per-cohort receipts:

Cohort	Classification	Latency (ns)	Source receipt
meta_v2_canary_default	REJECTED_VALIDATION	42,601,768,575 (42.60 s)	canary spark-open v2
meta_v2_canary_low_reasoning	COMPLETED	18,912,619,321 (18.91 s)	canary v2 low-reasoning
meta_v2_core_run_clean_happy_path	REJECTED_VALIDATION	22,446,966,812 (22.45 s)	core-run spark-open v1
meta_benchmark_96_mb96_000	REJECTED_VALIDATION	13,436,939,856 (13.44 s)	mb96-000 spark-open v1
meta_benchmark_384_mb384_005	REJECTED_VALIDATION	19,250,391,126 (19.25 s)	mb384-005 spark-open v1

Combined (n=5 receipts): min 13,436,939,856 ns; mean floor 23,329,737,138 ns; p50 19,250,391,126 ns; p95 42,601,768,575 ns; max 42,601,768,575 ns. The property meta experiment (LIVE_RECEIPT, n=1): latency 42,046,462,412 ns (42.05 s); tokens input 1,929 / output 1,365 / reasoning 157 / total 3,294 / cached input 0. The R11 final live validation: tokens prompt 1,822 / completion 1,164 / reasoning 493 / cached input 1,792.

Placeholders are NOT measurements (verbatim from the artifacts' own not_measured blocks): offline mock harness NOT_MEASURED (scripted latency 0.0); stage-1 packet run NOT_MEASURED (0.0 placeholders for failed calls); committed R11 records carry no latency field — never zero-filled. Every cohort above is a single observation (n=1): exact recorded values only; no interval is computable and none is emitted.

Targets are TARGETS, never claimed achievements: deterministic decision <= 3 s; validated assessment p50 <= 10 s / p95 <= 15 s. No live receipt measures a validated assessment end-to-end in under 15 seconds, so no sub-15-second latency claim is made (NOT_MEASURED). The aspirational "15-minute funding" timeline is likewise NOT_MEASURED — no receipt supports it, and it must never be presented as a measured capability. Offline engineering timings (seven-stage pipeline milliseconds per scenario; decide-path median 0.043 ms per client decision) are HOST-LABELED measurements that stay out of committed artifacts and are therefore not figures of this manuscript.

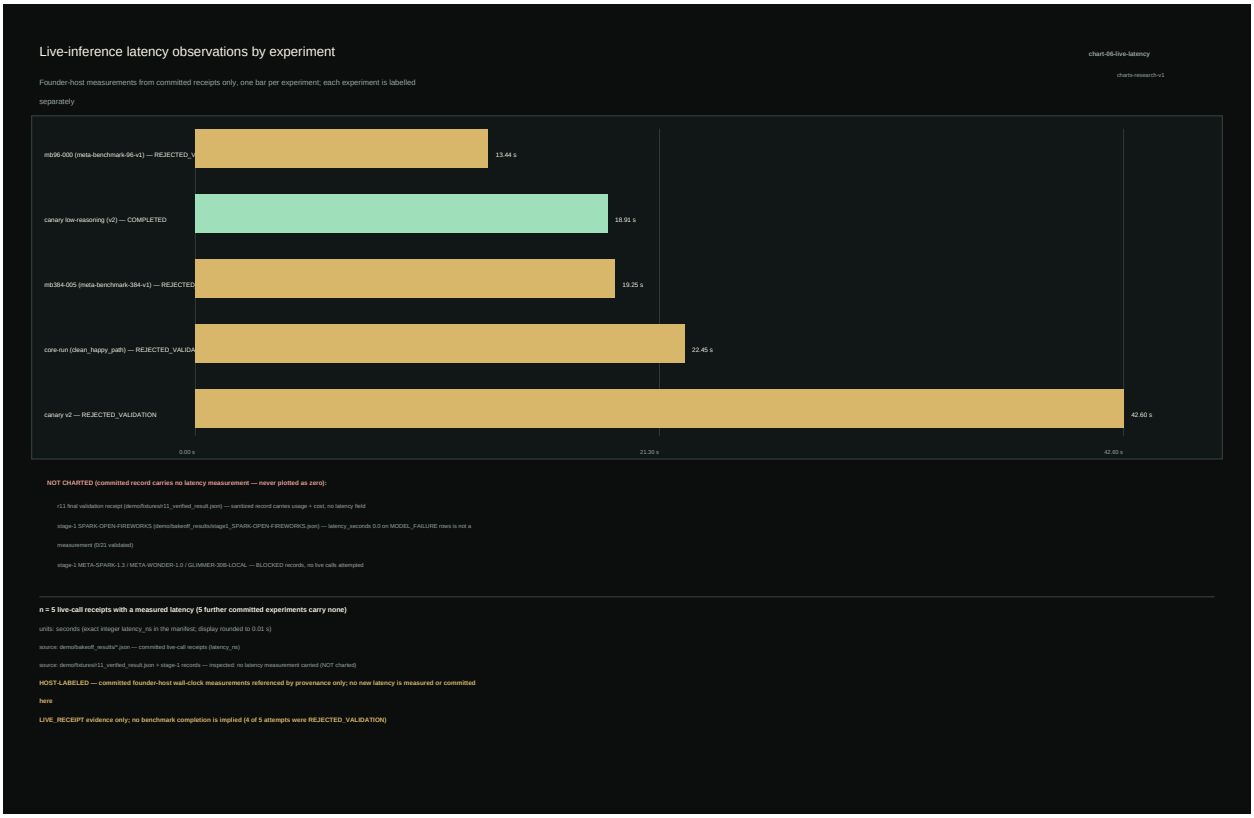


Figure 7. Live-inference latency observations by experiment — n = 5 live-call receipts with a measured latency (5 further committed experiments carry none). Units: seconds (exact integer latency_ns in the manifest; display rounded to 0.01 s). Sources: demo/bakeoff_results/*.json — committed live-call receipts (latency_ns); demo/fixtures/r11_verified_result.json + stage-1 records — inspected: no latency measurement carried (NOT charted).

4.6 Cost position and envelopes (integer micro-USD; ESTIMATED)

Committed-record cost items (per-item evidence label REAL_INFERENCE / LIVE_RECEIPT; every value ESTIMATED or NOT_STATED_IN_RECEIPT, never billed):

Item	micro-USD	Billing status
Stage-1 packet run (SPARK-OPEN-FIREWORKS)	54,993	NOT_STATED_IN_RECEIPT
Stage-1 diagnostic probe (1 call)	17,627	NOT_STATED_IN_RECEIPT
Meta-v2 canary (default reasoning)	12,459	ESTIMATED_FROM_PRICING_EVIDENCE
Meta-v2 canary (low reasoning)	5,635	ESTIMATED_FROM_PRICING_EVIDENCE
R11 final live validation	5,254	NOT_STATED_IN_RECEIPT
R11 first live call (superseded request document)	5,246	NOT_STATED_IN_RECEIPT
Meta-v2 core run (clean happy path)	5,416	ESTIMATED_FROM_PRICING_EVIDENCE
Meta-benchmark-96 (mb96-000)	7,487	ESTIMATED_FROM_PRICING_EVIDENCE
Meta-benchmark-384 (mb384-005)	6,620	ESTIMATED_FROM_PRICING_EVIDENCE

Committed-record totals (read back and re-summed in-process): **120,737 micro-USD with the diagnostic probe; 103,110 micro-USD without it** (USD 0.120737 / USD 0.103110). The property meta experiment receipt adds 8,213 micro-USD (ESTIMATED_FROM_PRICING_EVIDENCE), giving a cumulative budget

position over committed records of **128,950 micro-USD (USD 0.128950)**. `actual_billed_cost_usd` is null in every committed receipt; nothing here is presented as billed. Excluded honestly (recorded only in the private ledger, in no committed receipt): the executed meta-semantic-diagnostic attempt, the R11 case's UNAVAILABLE transport attempts, and the repeat-run R1/R2 receipts.

Planning estimates (OFFLINE_MOCK planning constants, NOT spend): spark-open 2,300,000; glimmer 6,800,000; repeated trials 3,000,000; total ready paths 11,800,000 micro-USD, within the hackathon allocation.

Cost envelopes for future live programs (PLANNING_ASSUMPTION-derived, computed from roster pricing, integer micro-USD, ESTIMATED). Per call: realistic 6,000 (spark-tier models) / 25,600 (muse_wonder_1_0); doubled input 7,750 / 31,200; absolute bound (output-capped) 12,375 / 55,600; observed runaway (4,096-token completion cap, measured once in the stage-1 diagnostic probe) 19,158 spark-tier / 87,520 wonder. Full-384 three-model roster totals: realistic 14,438,400; doubled input 17,932,800; absolute bound 30,854,400; runaway 48,321,024. The realistic tier (6,000/call) sits inside the measured 5,254–12,459 committed-estimate band, but the observed runaway behavior — every attempted packet specialist call in stage-1 ran to the completion cap — makes the absolute-bound tier mandatory in any spend plan.

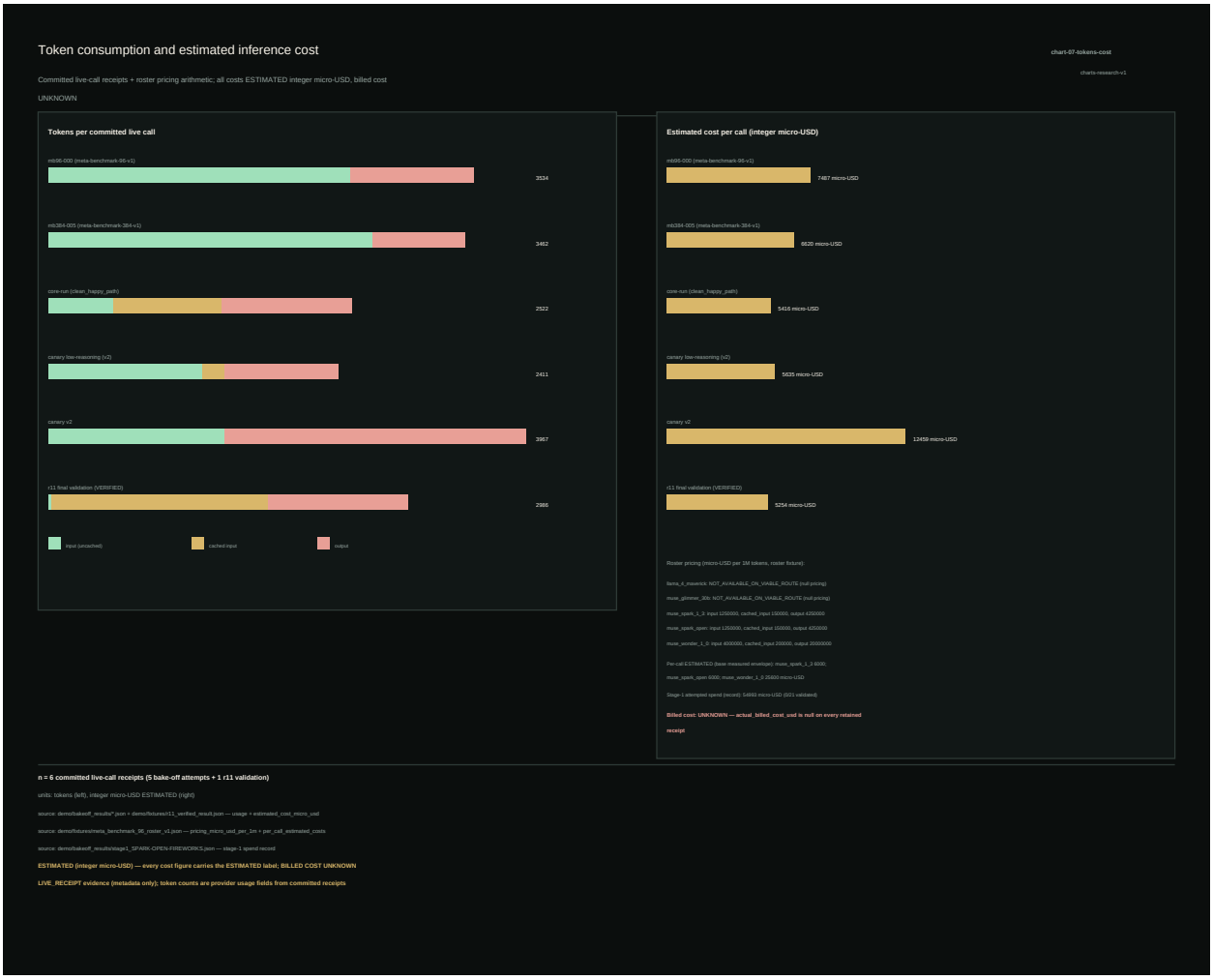


Figure 8. Token consumption and estimated inference cost — n = 6 committed live-call receipts (5 bake-off attempts + 1 r11 validation). Units: tokens (left), integer micro-USD ESTIMATED (right). Sources: demo/bakeoff_results/*.json + demo/fixtures/r11_verified_result.json — usage + estimated_cost_micro_usd; demo/fixtures/meta_benchmark_96_roster_v1.json — pricing_micro_usd_per_1m + per_call_estimated_costs; demo/bakeoff_results/stage1_SPARK-OPEN-FIREWORKS.json — stage-1 spend record.

4.7 Safety and failure attribution

Four frozen failure-attribution tiers (sprint03-failure-attribution-v1): MODEL_FAILURE, VALIDATOR_FAILURE, SCHEMA_FAILURE, EVIDENCE_FAILURE; attribution is UNDETERMINED where the response text was never retained (NO_FAILURE is the negative-control label, not a failure tier).

Attribution class	Records (computed)
MODEL_FAILURE	3
VALIDATOR_FAILURE	2
SCHEMA_FAILURE	1
EVIDENCE_FAILURE	1
UNDETERMINED	1
NO_FAILURE	1

Corpus: 8 records (OFFLINE_MOCK authored receipt corpus, computed via the frozen module read-only). Textless records: 3 of 8; textless records assigned a model-or-validator class: 0 — **no model defect is asserted from unretained text.**

Safety-violation rejections (LIVE_RECEIPT, read from the committed statistics fixture): 4 attempts, classification REJECTED_VALIDATION, error SAFETY_VIOLATION; quality claim NONE — a SAFETY_VIOLATION rejection with the response never retained proves nothing about model quality either way. The three unretained cases (core-run, mb96-000, mb384-005) carry attribution UNDETERMINED_FROM_RETAINED_EVIDENCE: the recorded filter-triggered class and the unchanged error code are reported — never a confirmed model defect, never a confirmed false positive. The failure taxonomy defines 14 classes with 0 recorded harness-validation failure records.

The frozen SAFEGUARDS tuple rides every assessment (verbatim): FICTIONAL_INPUTS_ONLY, ILLUSTRATIVE_NOT_CREDIT_ADVICE, HUMAN_REVIEW_REQUIRED, NO_CREDIT_APPROVAL, NO_ISSUANCE, NO_FUNDING, NO_LEGAL_OR_TITLE_VERIFICATION.

5. Statistical Adequacy of the Evidence Base

Source: the committed statistical inference and power report (`docs/THAL3S16-STATISTICAL-INFERENCE-AND-POWER-REPORT.md`) and its computation module (`demo/statistical_power_v1.py`), both digest-pinned; every figure below was read back and cross-verified by in-process recomputation for this manuscript. Planning rates are PLANNING_ASSUMPTION-labeled.

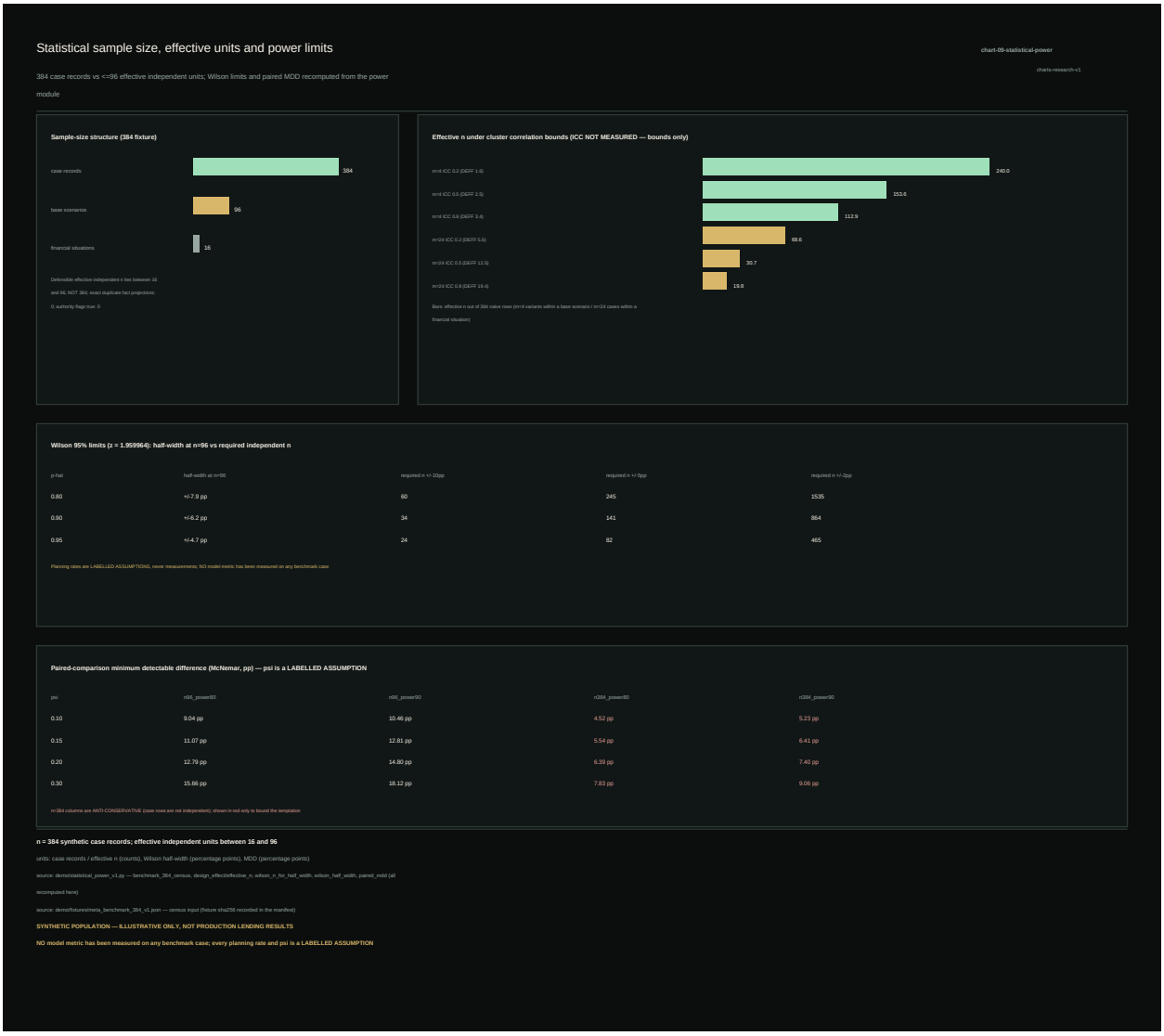


Figure 9. Statistical sample size, effective units and power limits — n = 384 synthetic case records; effective independent units between 16 and 96. Units: case records / effective n (counts), Wilson half-width (percentage points), MDD (percentage points). Sources: demo/statistical_power_v1.py — benchmark_384_census, design_effect/effective_n, wilson_n_for_half_width, wilson_half_width, paired_mdd (all recomputed here); demo/fixtures/meta_benchmark_384_v1.json — census input (fixture sha256 recorded in the manifest).

5.1 What the 384 cases are — and are not

The 384-case census is an authored balanced factorial design (Section 3.3), seed 20261031, byte-reproducible. It is NOT 384 completed live assessments (zero exist), NOT a representative sample of real UHNW clients (all values authored synthetic), and its variants are CORRELATED BY CONSTRUCTION (the four variants of each base scenario share subject, archetype and requested amount; the six regime rows of each financial situation share the subject). The blocked-majority outcome distribution (270 BLOCKED / 114 ILLUSTRATIVE_ONLY) is by design: two of the four variant archetypes are deliberately adverse. The only population these cases support inference about is the deterministic contract itself.

5.2 Effective independent sample size (computed bounds)

Intraclass correlation (ICC) is NOT MEASURED — no repeated live measurement exists at benchmark scale — so design effects are given as bounds (PLANNING_ASSUMPTION on ICC: 0.2 / 0.5 / 0.8):

Cluster model	ICC rho	DEFF	Effective n of 384
m=4 (variants within a base scenario)	0.2	1.6	240.0
m=4	0.5	2.5	153.6
m=4	0.8	3.4	112.9
m=24 (all cases of a financial situation)	0.2	5.6	68.6
m=24	0.5	12.5	30.7
m=24	0.8	19.4	19.8

Determination (S05 #2, read back verbatim in substance): the defensible effective independent sample size lies between **16** (financial situations, the only units not repeated along any axis) and **96** (base scenarios, distinct subject/regime cells) — NOT 384. Every interval computed on the raw 384 rows is anti-conservative: naive half-widths inflate by ×1.265 / ×1.581 / ×1.844 at the variant level (ICC 0.2 / 0.5 / 0.8) and by ×2.366 / ×3.536 / ×4.405 if regime repeats of the same situation also correlate. Cross-fixture pooling is capped at 384 unique situations (the 96 variant-0 cases are byte-twins of the 96-case packet).

5.3 Required independent case counts (Wilson 95%)

Required INDEPENDENT completed cases for a Wilson 95% interval at planning rate p-hat (z = 1.959964; rates are LABELLED ASSUMPTIONS):

Planning p-hat	Metric	±10 pp	±5 pp	±2 pp
0.80	grounding pass rate	60	245	1,535
0.90	completion rate	34	141	864
0.95	high-accuracy metrics	24	82	465
0.01	safety failure (±2.0 / ±1.0 / ±0.5 pp)	134	432	1,603
0.02	safety failure (±2.0 / ±1.0 / ±0.5 pp)	213	772	3,052

"Independent" means completed live cases from distinct base scenarios — against which the benchmark's carrying capacity is 96 scenarios, and no model metric has been measured on any of them. The safety rows show why per-case safety rates are essentially unmeasurable at benchmark scale: even 384 independent completed cases resolve a 1% rate only to about ±1 pp.

5.4 Paired-comparison power (matched cases)

Design: MATCHED CASES — the identical frozen case run under both models (temperature 0, single call, frozen validator); McNemar normal approximation on discordant pairs; psi (discordant-pair fraction) is a LABELLED ASSUMPTION. Minimum detectable differences at n=96 pairs:

psi	MDD at 80% power	MDD at 90% power
0.10	9.04 pp	10.46 pp
0.15	11.07 pp	12.81 pp
0.20	12.79 pp	14.80 pp
0.30	15.66 pp	18.12 pp

Required completed PAIRS for a target difference: Delta=10 pp at psi=0.15 -> 118 pairs (80% power) / 158 (90%); at psi=0.10 -> 79 / 106; psi=0.20 -> 157 / 211; psi=0.30 -> 236 / 316. Delta=5 pp at psi=0.15 -> 471 / 631 (314 / 421 at psi=0.10; 628 / 841 at psi=0.20; 942 / 1,261 at psi=0.30). The n=384-row variant of this arithmetic is anti-conservative and rejected. Exact-test floor: a fully one-sided discordant split needs >= 6 discordant pairs for two-sided $p < 0.05$ ($p = 0.03125$ at 6:0), rising to 7 under Bonferroni control over 3 comparisons. Reference unpaired planning: 199 per arm to detect 0.90 vs 0.80 at 80% power; 141 per arm for 0.95 vs 0.85.

Determination: a paired comparison at the benchmark's 96-scenario capacity detects only ~9–18 pp differences at 80–90% power; ranking two models whose true rates differ by less than 10 pp requires ~118–158 completed paired cases at psi=0.15 — more than the benchmark's capacity, and none of it has been measured live.

5.5 Per-regime adequacy

Each of the six regimes carries 64 case rows but only **16 defensible independent units** — 26.7% of the 60 required for a ±10 pp interval at p-hat=0.80. Sixteen independent observations buy a Wilson half-width of ±18.21 pp at p-hat=0.80 (±16.26 pp at p-hat=0.90) — descriptive only. The per-regime expected-outcome distribution is identical in all six regimes (19 ILLUSTRATIVE_ONLY / 45 BLOCKED) by construction, so the regime axis supplies macro context only and changes no deterministic engine outcome.

Determination: per-regime interval estimation is NOT ADEQUATE at benchmark scale; per-regime results must stay descriptive; a per-regime rate with a 95% interval needs ~60 completed independent live cases per regime (~360 across six regimes) versus zero completed today.

5.6 Rare safety-failure detection limits

Zero-failure upper bounds (computed):

Completed independent n	Rule of three 3/n	Wilson 95% upper (0 failures)
48 (stage-48 scale)	6.25%	7.41%
64 (one regime's rows)	4.69%	5.66%
96 (base scenarios)	3.13%	3.85%
384 (all rows, anti-conservative)	0.78%	0.99%

Completed calls needed to bound the true safety-failure rate with zero observed failures: <=5% -> 60 (rule of three) / 73 (Wilson); <=1% -> 300 / 381; <=0.5% -> 600 / 765; <=0.1% -> 3,000 / 3,838. Detection probability of at least one failure: at a true 1% rate, 61.90% with n=96 and 97.89% with n=384; at a true 0.1% rate, 9.16% with n=96 and 31.90% with n=384 — a clean 384-case run is NOT a meaningful probe for rates at or below 0.1%.

5.7 Live evidence position and honest intervals

Read back from the committed records and cross-verified by recomputation: inference completion 3 of 9 attempted = **33.33%**, Wilson 95% interval **[0.1206, 0.6458]**; validated assessment success 3 of 3 = 100%, Wilson 95% interval **[0.4385, 1.0]** (an interval, not a claim of perfection). Live safety: zero failures observed on n = 3 completed calls, Wilson 95% upper bound **0.5615** (displayed in the committed artifact as 56.2%) — live safety behavior is essentially UNMEASURED; bounding it at even 5% requires 73 completed independent live calls. Five recorded SAFETY_VIOLATION incidents exist on NOT_SCORED rejected attempts (incident counts only; no failure rate is computed from rejected attempts anywhere).

Minimum vs preferred live samples (PLANNING values): minimum for any interval at all: n >= 2 completed independent cases; for a ±10 pp interval on completion/grounding rates: 60 (34 at p-hat=0.90); preferred for a per-model rate with a usable interval: >= 245 (~ 2.6× the benchmark's 96-scenario capacity — requiring case generation beyond the current fixture, a founder decision); minimum for a paired comparison: 118 completed pairs (158 at 90% power; 471 for 5 pp); minimum for a 5% safety bound: 73 (381 for 1%).

Determination: the live evidence base (3 completed single cases; 0 on the benchmark) supports NO inferential statement about any model. Every minimum above is unmet by one to two orders of magnitude.

6. Model Comparison Readiness

The comparison-preparation module (`demo/meta_comparison_prep_v1.py`) exists so that a future head-to-head evaluation cannot improvise its own success criteria. Its properties, verified in-process for this manuscript:

- **Honest roster:** five models with fixture-verbatim availability tiers;

only muse_spark_open is EXECUTION_PATH_TESTED; the other four have zero live calls. Per-model buckets are computed from receipts and used-markers only.

- **Matched-case plan:** a paired-by-case-id plan over identical frozen

case sets per model (pairing key = the fixture's own case ids), with the unit pin enforced: N transport receipts under one experiment/scenario id count as ONE case; two contract variants of one scenario count as TWO cases.

- **Six separated dimensions** (financial grounding, safety, latency, cost,

abstention, completed throughput), each with its own metric and UNKNOWN propagation: a dimension that is unmeasured renders NOT_MEASURED (null, never 0) and the publish gate refuses — PUBLISH_REFUSED_UNMEASURED_DIMENSION. No blended score exists (blending raises by design).

- **Winner evidence bar (pre-registered, committed before any live

batch):** paired design on identical case ids; the minimum completed paired sample from Section 5.4; all six dimensions measured; only LIVE_RECEIPT results admissible; pre-registered per-dimension thresholds; multiplicity control (exact McNemar plus seeded paired bootstrap, >= 6 discordant pairs, 7 under 3-comparison family-wise control); no blending. The bar's current status: NOT MET — by one to two orders of magnitude.

- **NO_WINNER_DECLARED:** the declaration function raises

InsufficientEvidenceError unconditionally (exercised in-process for this manuscript), all ten role-specific winner slots read PENDING_LIVE_BAKEOFF, and every fit score is UNKNOWN (NOT_MEASURED), never zero.

Evidence: no real Meta model has been scored under the representative benchmark; every comparative number in the repository is OFFLINE_MOCK (scripted machinery validation) or SYNTHETIC (authored deterministic expectations). **Determination:** NO model-versus-model quality ranking is supported by live evidence; zero paired live observations exist. All existing harness observations (12 comparable pairs; recognition, sensitivity, calibration and assumption-as-fact rates all 0.0 over the scripted provider) validate machinery over a mock, and one honest harness observation stands: the v1 scripted provider does not reference scenario.* facts, so its measured sensitivity is None — an observation about the harness, not a model claim.

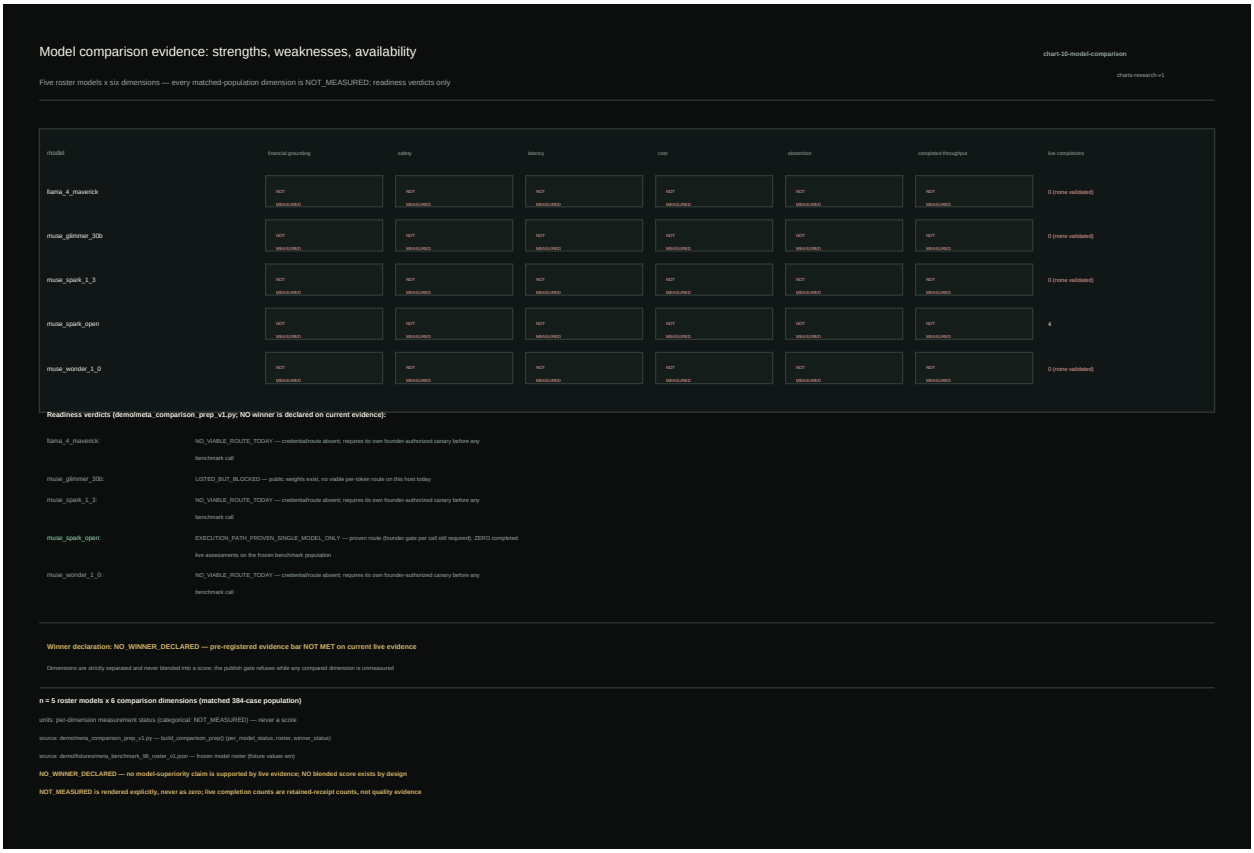


Figure 10. Model comparison evidence: strengths, weaknesses, availability — n = 5 roster models x 6 comparison dimensions (matched 384-case population). Units: per-dimension measurement status (categorical: NOT_MEASURED) — never a score. Sources: demo/meta_comparison_prep_v1.py — build_comparison_prep() (per_model_status, roster, winner_status); demo/fixtures/meta_benchmark_96_roster_v1.json — frozen model roster (fixture values win).

7. Limitations and Honest-Statement Registry

1. **Model quality on any benchmark population: NOT_MEASURED.** Zero completed live benchmark cases exist; no rate, interval or comparison exists or is implied.
2. **Live safety behavior: essentially UNMEASURED.** Three completed live calls, zero failures; 95% upper bound 56.2% (computed); bounding at 5% needs 73 completed calls.

3. **Actual billed costs: UNKNOWN.** Every cost figure is

ESTIMATED_FROM_PRICING_EVIDENCE or NOT_STATED_IN_RECEIPT (integer micro-USD); no billed amount is recorded anywhere.

4. **ICC of the variant/regime clusters: NOT_MEASURED.** Design effects are

bounded scenarios (0.2/0.5/0.8) over an authored population.

5. **Repeatability: NOT_MEASURED.** Two of three planned same-case calls

recorded (one completed, one rejected); pairwise agreement unmeasured; temperature-0 determinism is a per-run measured property, never assumed.

6. **Per-regime interval estimation: NOT ADEQUATE at benchmark scale**

(16 independent units per regime vs 60 required).

7. **The two approvals without receipts: UNDETERMINED** from metadata alone

(INCOMPLETE bucket; existence counted only, contents not read).

8. **Confidential policy sources are not re-readable in-band.** Threshold

provenance relies on the committed audited matrix and its recorded anchor verification (57 of 57 anchors verified at audit time); the matrix is an audit-time v1 snapshot beside a live v2 engine (a recorded, test-pinned property, not a discrepancy).

9. **Single-observation receipts.** Every live latency, token and grounding

figure rides an n=1 receipt: exact recorded values, no intervals. 10. **RECORDED bucket population: UNVERIFIED** as a separate committed entry in the S05 read-first set; recorded grading outcomes ride on LIVE_RECEIPT records. 11. **The recovered frontend is a reconstruction from publicly served bytes** (provenance committed); equality with the original development source is unverifiable from this instance and was not attempted. 12. **No funding-timeline claim.** No receipt supports any specific end-to-end funding timeline; the aspirational sub-15-second assessment target and any faster funding narrative remain NOT_MEASURED and must never be presented as measured capabilities.

8. Proposed Next Steps (all labeled Proposed: — none is an executed result)

- **Proposed:** a founder-authorized live paired evaluation of at least 118

completed matched cases (the pre-registered 10 pp / psi=0.15 / 80% power design), run under the frozen matched-case plan and the winner evidence bar as committed — the smallest design that could support any comparative statement.

- **Proposed:** a new versioned benchmark fixture (>= 245 independent

situations) if a per-model rate with a usable ±5 pp interval is wanted; this exceeds the current fixture's 96-scenario capacity and is a founder decision.

- **Proposed:** a dedicated safety run of 73 completed independent calls to

put a 5% zero-failure bound on live safety behavior (381 calls for 1%).

- **Proposed:** surfacing the blocking reason code in the demonstration

frontend's final card for blocked clients (currently the block is explicit — $0 / BLOCKED — but the machine reason code lives only in the judge-flow payload); a design change, so it requires founder approval under the no-redesign contract.

- **Hypothesis:** the observed stage-1 runaway behavior (every attempted

packet specialist call reaching the 4,096-token completion cap) is driven by prompt-shape (packet-style context) rather than model choice; untested — any live program should carry the absolute-bound cost tier until this is measured.

Appendix A — Machine-Readable Identifier Register

(Exact tokens retained for auditability; the narrative uses the readable labels mapped in the Style Key.)

Engine and contracts: `credit-engine-v2`; `credit-request-v1`; `ltv-rounding-contract-v2` (convention: ceiling; founder approval status: APPROVED); decision output vocabulary `ILLUSTRATIVE_CAPACITY` / `REVIEW_REQUIRED` / `BLOCKED`; fixture schemas `clients30-v1`, `meta-benchmark-96-v1`, `meta-benchmark-384-v1`, `meta-benchmark-96-roster-v1`, `economic-scenario-results-v1`, `policy-coverage-matrix-v1`; judge-flow schema `r12-judge-flow-v1`; failure attribution contract `sprint03-failure-attribution-v1`; comparison schemas `meta-comparison-prep-v1` / `meta-comparison-matched-plan-v1`.

Reason codes (30-client census, recomputed): `BLOCKED_REQUEST_EXCEEDS_CAPACITY` (6), `BLOCKED_OWNERSHIP_OR_LIEN_EVIDENCE` (3), `BLOCKED_INSUFFICIENT_PROPERTY_EVIDENCE` (2), `BLOCKED_INELIGIBLE_COLLATERAL` (1), `BLOCKED_MISSING_FINANCIAL_EVIDENCE` (1), `REVIEW_INSUFFICIENT_CASHFLOW` (5), `REVIEW_RISK_CONCENTRATION` (2), `REVIEW_SEVERE_SINGLE_DIMENSION` (1). Benchmark blocking reasons: `REQUEST_EXCEEDS_CAPACITY`, `STALE_EVIDENCE`, `VALUE_CONFLICT`.

Client and benchmark case ids: demonstration clients `cf30-001` through `cf30-030` (30 records, categories A–E); benchmark case ids `mb384-000` through `mb384-383` (384 records) and the 96-case packet's `mb96-*` ids; variant archetypes `reference_authored_baseline`, `fresh_evidence_low_leverage`, `stale_evidence_wider_disagreement`, `conflicting_fresh_high_leverage`; regime identifiers `normal`, `high_interest_rates`, `property_downturn`, `liquidity_crisis`, `recession`, `recovery` (13-regime scenario library: `baseline_normal`, `credit_liquidity_crunch`, `high_inflation`, `higher_interest_rates`, `insurance_climate_shock`, `market_recovery`, `rapid_rate_cuts`, `real_estate_price_decline`, `recession`, `regional_ca_downturn`, `severe_luxury_correction`, `strong_appreciation`, `unemployment_shock`).

Engine constants (illustrative demo parameters, PROPOSED_UNCALIBRATED / PENDING_POLICY_SOURCE): `DEMO_ILLUSTRATIVE_LTV_BPS` 4000; `DEMO_WORKED_VALUE_CENTS` 500000000; `DEMO_WORKED_REQUESTED_CENTS` 150000000; `DEMO_STALENESS_HAIRCUT_BPS` 2500; haircut cap 1500 bps; `DEMO_RISK_CONCENTRATION_HIGH_DIMENSIONS` 3. v2 calibration parameter `P2-concentration-response-review-branch`, review reason code `REVIEW_RISK_CONCENTRATION`, policy refs `POL-A-027`, `POL-B-019`, `POL-B-022`; composite-score guard `POL-A-025`. Historical pre-calibration reason code: `BLOCKED_RISK_CONCENTRATION`.

Policy matrix status vocabulary: `IMPLEMENTED_GROUNDED`, `PARTIALLY_IMPLEMENTED`, `PROPOSED_UNCALIBRATED`, `NO_MAPPING`; maturity labels `GROUNDED`, `GROUNDED_SIGNAL_PROPOSED_RESPONSE`, `DRAFT_INTERNAL_NOT_PRODUCTION_APPROVED`, `NOT_APPLICABLE`; vocabulary status `PENDING_POLICY_SOURCE`; out-of-matrix refs render `policy-ref:PENDING_SOURCE`. Data label: `DERIVED_FROM_FOUNDER_CONFIDENTIAL_POLICY_PRIVATE_REPO_ONLY`.

Live registers (case keys, verbatim): `r11-single-call:valid`; `meta_v2_canary_default`; `meta_v2_canary_low_reasoning`; `meta_v2_core_run:clean_happy_path`; `meta_semantic_diagnostic:clean_happy_path`; `meta_benchmark_96:mb96-000`; `meta_benchmark_384:mb384-005`; stage-1 keys `stage1:clean_happy_path`, `stage1:consistent_evidence_spread_100`, `stage1:mild_valuation_disagreement_20bps`. Property meta experiment: `meta-property-c6fb6071-v1`. Repeatability experiment: `meta-property-repeat-c6fb6071-v1`.

Roster model ids and tiers: `muse_spark_open` (EXECUTION_PATH_TESTED, Fireworks-hosted route; LIVE_PROVEN_R11_VERIFIED_9_OF_9); `muse_spark_1_3` (AVAILABLE_NOT_LIVE_TESTED_BY_THAL3S); `muse_wonder_1_0` (AVAILABLE_PREVIEW_NOT_LIVE_TESTED_BY_THAL3S); `muse_glimmer_30b` (BENCHMARK_PACKET_READY_NOT_LIVE_TESTED); `llama_4_maverick` (not available on a viable route).

Evidence-bucket vocabulary (verbatim): LIVE_RECEIPT, OFFLINE_MOCK, RECORDED, SYNTHETIC, INCOMPLETE; legacy mapping REAL_INFERENCE == LIVE_RECEIPT. Honesty vocabulary: NOT_MEASURED, UNKNOWN, UNDETERMINED, PLANNING_ASSUMPTION, DRAFT_NOT_PRODUCTION_APPROVED, PROPOSED_UNCALIBRATED, PENDING_POLICY_SOURCE, NOT_APPLICABLE, SYNTHETIC_DETERMINISTIC, DETERMINISTIC_EXPECTATIONS, SEMANTIC_GROUNDING_NOT_ESTABLISHED, NO_WINNER_DECLARED, PUBLISH_REFUSED_UNMEASURED_DIMENSION, PENDING_LIVE_BAKEOFF.

Safeguards tuple (frozen, verbatim): FICTIONAL_INPUTS_ONLY, ILLUSTRATIVE_NOT_CREDIT_ADVICE, HUMAN_REVIEW_REQUIRED, NO_CREDIT_APPROVAL, NO_ISSUANCE, NO_FUNDING, NO_LEGAL_OR_TITLE_VERIFICATION.

Appendix B — Provenance Register (source commits)

Source	Artifact	Source commit
Frontend recovery provenance	`demo/property_preview/RECOVERY_README.md` + pinned-diff test	`2e8b00f11dd0b76d7f1d2b5431156160ac9f1eba`
Recorded browser QA	`docs/SPRINTORIG_S07_REPORT.md`	`0318769c763dbee8295994a9c239d87e94495812`
30-client outcomes + v2 calibration	`demo/fixtures/clients_30_v1.json`	`653e0aa867fc4ca372959fd488547ee72d627759`
Policy coverage audit	`demo/fixtures/policy_coverage_matrix_v1.json`	`d6f5b03f3a14ae1922923aed695f51dfa49cd639`
Statistical adequacy	S05 report + module	`1036fce059e739135547a572f38a1f25f3581ad8`
Comparison readiness	`demo/meta_comparison_prep_v1.py`	`7176963326200988b111444bc7377fc83d46db59`
V8 report base	V8 generator integration head	`d32d4d8b5be36e22b4a5353338415924da8bdf3a`

Appendix C — Digest Register (sha256, recomputed at authoring)

Artifact	sha256
demo/fixtures/clients_30_v1.json	80a299240ff3608e34d9d99f4f521c5fbfcfc36abb55e042c57e7be8c26cce85
demo/fixtures/policy_coverage_matrix_v1.json	2d38079e91440d9d9f7f5b8ca368c6b549e46b253d9995f2e9b387f9e854041b
demo/fixtures/meta_benchmark_384_v1.json	dbfd874b93090f0c626e60195d128b52afc48cef19ce4e63e86e017a493bd01f
demo/fixtures/meta_benchmark_96_roster_v1.json	3f65005ebd91ebd5cfa0835fef317292859348ee1d5d1b4be84118d9202695a5
docs/THAL3S16-STATISTICAL-INFERENCE-AND-POWER-REPORT.md	4233d1b729598fe5ccab2cb8f9556788b1fafb23b4d4389d4336e33fea9bb5ac
demo/statistical_power_v1.py	e06cf8d9b5a0e54a0a432ccb80bfdb4c47860f6687a3d2f759c259553c376097
demo/meta_comparison_prep_v1.py	818b05f9da631726de1df4a3a113742725bf72a6b59e7b9c3aa048542a1c6a6b
docs/SPRINTORIG_S07_REPORT.md	822e7af31f3cadde800ae6110d65a924e9928ea0178b0af4da45e89ce5b77dad
demo/property_preview/clients30-integration.js	970bf408942a8388f3da92d8d190a298a514aadfc29e0639d80db3d46810c94d
THAL3S16_Meta_Model_Financial_Intelligence_Report_V8.pdf	a0076b9dfe7709b390ef093896188aa919c1fff9d2ec82cec2e148180ebc5414

Appendix D — Procedural Commentary (sprint mechanics; kept out of the narrative)

- This manuscript is the SprintVis S04 deliverable (Research Editorial

Rewrite), authored in an isolated worktree on the sprint-orig integration base. The narrative source is the committed V8 report content plus the committed sprint reports; figures were re-read and recomputed through `demo/clients30.py`, `demo/credit_engine.py` and `demo/statistical_power_v1.py`, and are anchored by `demo/tests/test_sprintvis_s04_manuscript.py`.

- The V8 report itself is additive over the frozen V1–V7 PDFs, which remain

byte-identical and pinned by sha256 in their suites; the V8 PDF is timestamp-free and byte-reproducible (13 pages; single pinned date literal `D:20261101080000Z`).

- Sprint-orig specialist reports S01–S08 (frontend markers, policy and

financial integrity, compact reasoning, ATTOM evidence normalization, statistical adequacy, matched-case comparison, browser QA, V8 integration) are the committed sub-records behind Sections 3–6; their suite counts and per-check verdicts live in those reports, not here.

- Frontend integration discipline: the property-intelligence demo's

changes are one clearly marked additive block plus one script tag; `property-intelligence.original.js`, the generated integration module, `judge-flow.json` and the coordinator-owned recovery test are untouched by this manuscript work.

- Suite requirements note: the manuscript suite is pure stdlib and offline;

it requires no PDF toolchain (it anchors text, not PDF bytes).

Appendix E — Glossary for Non-Engineering Readers

- **Basis point (bps):** one hundredth of a percentage point. 4000 bps = 40%. "pp" is whole percentage points.
- **Indicative capacity:** a number the engine computes to illustrate what the evidence could support — it is not an offer, approval or commitment.
- **Deterministic:** same inputs always produce the same outputs; the pipeline is a fixed calculation, not a judgment call.
- **Fail-closed:** when evidence is missing or contradictory, the system blocks or escalates to a human; it never guesses.
- **Evidence bucket:** the label (LIVE_RECEIPT, OFFLINE_MOCK, RECORDED, SYNTHETIC, INCOMPLETE) that says what kind of thing a number is; figures from different buckets are never mixed.
- **Wilson interval:** a statistically honest way to state "the true rate is between X and Y, with 95% confidence" — especially at small samples where naive percentages mislead.
- **Minimum detectable difference (MDD):** the smallest true gap between two systems an experiment could reliably detect; if the MDD is 11 pp, differences smaller than that are invisible to the design.
- **Meta model:** an external language model evaluated for evidence reading under strict validation; it never decides, approves or funds.
- **NOT_MEASURED / UNKNOWN / UNDETERMINED:** explicit labels that a value does not exist; they are never rendered as zero and never upgraded into claims.

End of manuscript. This document carries no claims beyond its committed records: no significance claim, no model-versus-model quality ranking, no autonomous-lending capability, no liquidity commitment. What is unmeasured is labeled unmeasured.

End of manuscript. This document carries no claims beyond its committed records: no significance claim, no model-versus-model quality ranking, no autonomous-lending capability, no liquidity commitment. What is unmeasured is labeled unmeasured.